# MAST30027: Modern Applied Statistics

Assignment 2, 2026 — Great Barrier Reef Consulting Report: Sample Solution

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**About this solution**

This is one defensible analysis. Other reports may receive full marks if their modelling decisions are statistically justified, their calculations are correct, and their conclusions follow from the evidence.

## Marking scheme

Each part is marked independently using the following common scale.

| Mark | Standard |
|---|---|
| **0** | No relevant working or response. |
| **1** | Minimal relevant working; major omissions or misconceptions; no defensible conclusion. |
| **2** | Substantial and generally appropriate working, but with errors, omissions, or unclear justification or interpretation. |
| **3** | Correct and complete working, with clear justification, interpretation, and communication appropriate to the audience. |

## Part 1: Executive summary

We analysed 2012 AIMS coral trout counts from 120 monitoring sites. Each count represents the same survey effort: five transects per site. We compared three broad latitude bands and two fishing-zone categories using count regression.

The initial Poisson analysis appeared to suggest that the difference between no-take and open sites varied by latitude band. However, the data were much more variable than a Poisson model allows. After correcting for this overdispersion with a quasi-Poisson model, there was insufficient evidence that the zone comparison differed among latitude bands.

The final model estimates that, after accounting for latitude band, the expected count at a no-take site is approximately **1.76 times** the expected count at an open site. The 95% confidence interval is approximately **1.25 to 2.49**. Expected counts were also substantially higher in the Southern band than in the Northern and Central bands.

One Southern no-take site had an unusually large count and was influential. Excluding it reduced the estimated no-take-to-open ratio, but the direction of the zone association and the need to allow for overdispersion remained. There was no evidence that it was a recording error, so it was retained in the primary analysis. One importance limitation of our study is that these are observational data from one year, so the results describe associations and do not establish that zoning caused the count differences.

## Part 2: Problem and data understanding

We aim to model coral trout counts based on latitude band and fishing status (open to fishing or located in a no-take zone), and investigate whether these variables and their interaction are statistically significant in the model.

The response is the coral trout count over five transect of the same length. There are two explanatory variables. One gives the reef latitude, summarised using a three-level categorical variable (Northern, Central, and Southern), and the other is a two-level categorical variable indicating whether the site is open to fishing (Open) or in a no-take zone (No-take).

We set `Northern` and `Open` as reference levels so that coefficients and expected-count ratios have convenient interpretations.

```
reef <- read.csv("aims_coral_trout_sites_2012.csv")

reef$latitude_band <- factor(
  reef$latitude_band,
  levels = c("Northern", "Central", "Southern")
)
reef$zone <- factor(
  reef$zone,
  levels = c("Open", "No-take")
)

dim(reef)
```

```
[1] 120   4
```

```
str(reef)
```

```
'data.frame':   120 obs. of  4 variables:
 $ site         : chr  "AIMS001" "AIMS002" "AIMS003" "AIMS004" ...
 $ latitude_band: Factor w/ 3 levels "Northern","Central",..: 3 3 3
3 3 3 3 3 3 3 ...
 $ zone         : Factor w/ 2 levels "Open","No-take": 2 2 2 2 2 2
2 2 2 2 ...
 $ coral_trout  : int  5 9 3 13 0 5 13 19 5 24 ...
```

```
# Number of sites in each latitude-band-by-zone group
with(reef, table(latitude_band, zone))
```

```
             zone
latitude_band Open No-take
     Northern   18      18
     Central    18      18
     Southern   24      24
```

```
# Check missing values (the data documentation states five transect
colSums(is.na(reef))
```

```
         site latitude_band          zone   coral_trout
            0             0             0             0
```

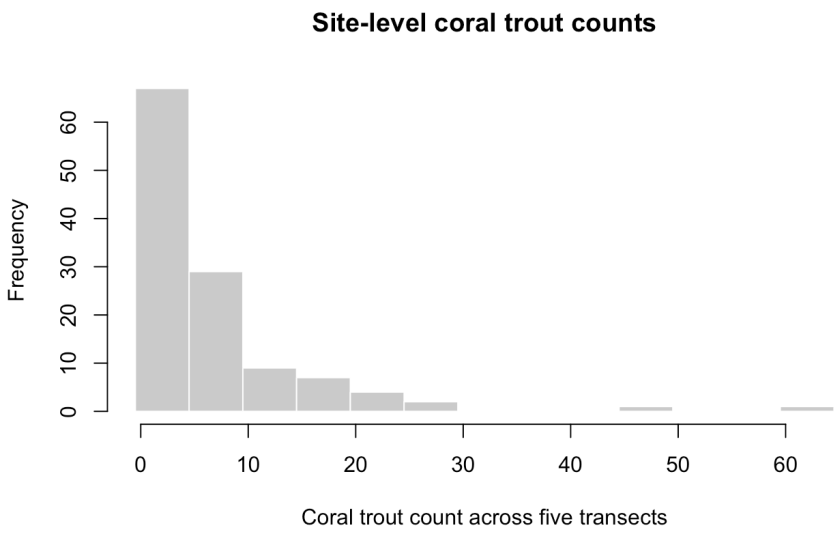
There are 18 sites in each zone in the Northern and Central bands and 24 in each zone in the Southern band. There are no missing values, and each row represents the same five-transect survey effort.

## Part 3: Exploratory analysis

```r
# Numerical summaries, using the same base-R approach as in class
with(
  reef,
  aggregate(
    coral_trout,
    list(latitude_band = latitude_band, zone = zone),
    function(x) c(
      n = length(x), mean = mean(x), sd = sd(x),
      median = median(x), max = max(x)
    )
  )
)
```

```
  latitude_band    zone       x.n    x.mean      x.sd  x.median
x.max
1      Northern    Open 18.000000  2.500000  1.947849  2.000000
6.000000
2       Central    Open 18.000000  2.722222  2.468759  2.000000
8.000000
3      Southern    Open 24.000000  7.708333  6.629835  5.500000
21.000000
4      Northern No-take 18.000000  3.388889  3.942810  1.500000
15.000000
5       Central No-take 18.000000  3.055556  2.775358  3.000000
12.000000
6      Southern No-take 24.000000 15.625000 14.209037 12.000000
61.000000
```

```r
hist(
  reef$coral_trout,
  breaks = seq(-0.5, max(reef$coral_trout) + 4.5, by = 5),
  col = "grey80",
  border = "white",
  xlab = "Coral trout count across five transects",
  main = "Site-level coral trout counts"
)
```




Distribution of site-level coral trout counts in the 2012 teaching extract.

```r
set.seed(30027)

x <- interaction(
```

```
  reef$latitude_band,
  reef$zone,
  lex.order = TRUE
)

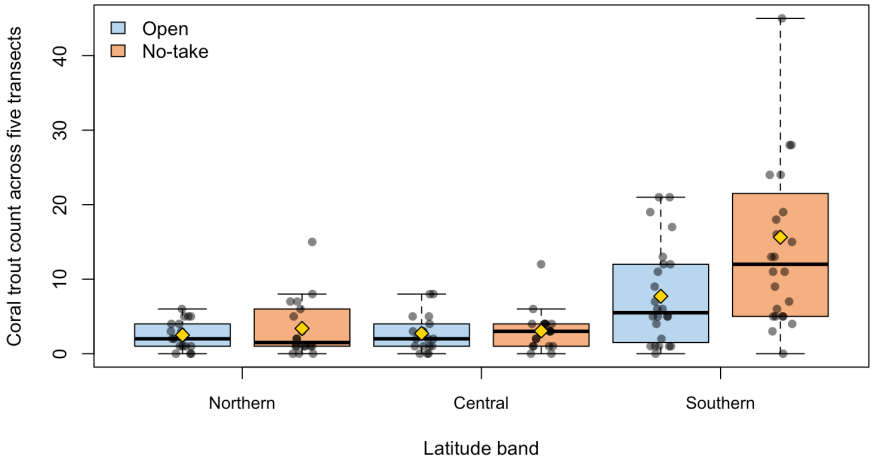
boxplot(
  coral_trout ~ x,
  data = reef,
  outline = FALSE,
  col = rep(c("#b8d8f0", "#f4b183"), 3),
  xaxt = "n",
  xlab = "Latitude band",
  ylab = "Coral trout count across five transects"
)

stripchart(
  coral_trout ~ x,
  data = reef,
  method = "jitter",
  vertical = TRUE,
  add = TRUE,
  pch = 16,
  col = adjustcolor("black", alpha.f = 0.55)
)

axis(
  1,
  at = c(1.5, 3.5, 5.5),
  labels = levels(reef$latitude_band),
  cex.axis = 0.9
)

points(
  1:6,
  tapply(reef$coral_trout, x, mean),
  pch = 23,
  bg = "gold",
  cex = 1.3
)

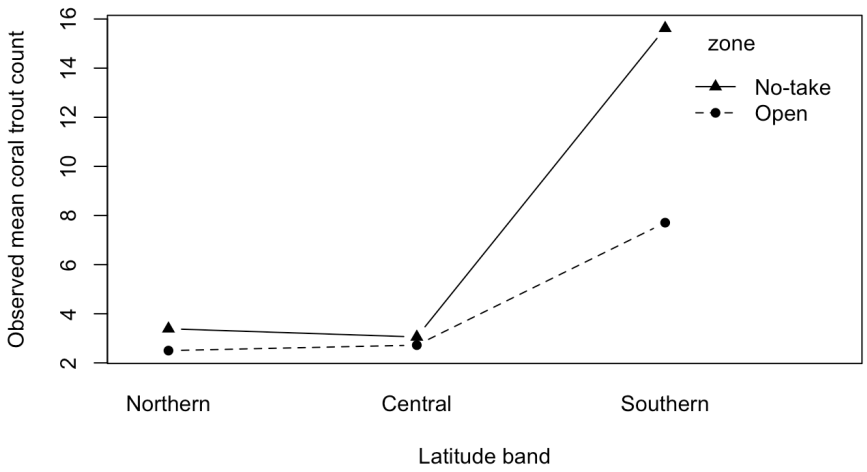
# Legend for zone
legend(
  "topleft",
  legend = levels(reef$zone),
  fill = c("#b8d8f0", "#f4b183"),
  border = "black",
  bty = "n"
)
```




Observed coral trout counts and group means by latitude band and zone. Points are horizontally jittered; diamonds show group means.

The counts are strongly right-skewed. Counts tend to be higher in the Southern band, particularly at no-take sites. The group means suggest a possible latitude-band-by-zone interaction: the no-take/open difference appears larger in the Southern band. There is also substantial within-group variability and an unusually large Southern no-take count.

```
with(
  reef,
  interaction.plot(
    latitude_band, zone, coral_trout,
    type = "b", pch = c(16, 17),
    xlab = "Latitude band",
    ylab = "Observed mean coral trout count"
  )
)
```




Observed mean site-level coral trout count by latitude band and zone.

An interaction plot provides a more direct comparison of group means. Non-parallel lines suggest that an interaction should be considered, but this plot does not account for the large within-group variability.

## Part 4: Initial GLM choice and formulation

Let $Y_i$ be the coral trout count at site $i$, let $B_i$ denote latitude band, and let $Z_i$ denote zone. The initial model is

$$Y_i \sim \text{Poisson}(\mu_i),$$

$$\log(\mu_i) = \beta_0 + \beta_{B,C} I(B_i = \text{Central}) + \beta_{B,S} I(B_i = \text{Southern}) + \beta_Z I(Z_i = \text{No-take}) + \beta_{CZ} I(B_i = \text{Central}) I(Z_i = \text{No-take}) + \beta_{SZ} I(B_i = \text{Southern}) I(Z_i = \text{No-take}).$$

Poisson regression is a reasonable starting point because the response is a count. In a Poisson regression, the variance function is $v(\mu) = \mu$. We also assume the observations (i.e., site-level observations) are independent. The log link ensures positive fitted means and makes exponentiated coefficients interpretable as ratios of expected counts.

```
poisson_full <- glm(
  coral_trout ~ latitude_band * zone,
  family = poisson,
  data = reef
)

summary(poisson_full)
```

```
Call:
glm(formula = coral_trout ~ latitude_band * zone, family = poisson,
    data = reef)

Coefficients:
                                  Estimate Std. Error z value
Pr(>|z|)
(Intercept)                        0.91629    0.14907   6.147
7.91e-10 ***
latitude_bandCentral               0.08516    0.20647   0.412
0.6800
latitude_bandSouthern              1.12601    0.16622   6.774
1.25e-11 ***
zoneNo-take                        0.30421    0.19651   1.548
0.1216
latitude_bandCentral:zoneNo-take  -0.18870    0.27786  -0.679
0.4971
latitude_bandSouthern:zoneNo-take  0.40236    0.21607   1.862
0.0626 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 977.60  on 119  degrees of freedom
Residual deviance: 570.62  on 114  degrees of freedom
AIC: 939.3

Number of Fisher Scoring iterations: 5
```

$$\log(\mu_i) = 0.91629 + 0.08516 \times I(B_i = \text{Central}) + 1.12601 \times I(B_i = \text{Southern}) + 0.30421 \times I(Z_i = \text{No-take}) + (-0.18870) \times I(B_i = \text{Central}) I(Z_i = \text{No-take}) + 0.40236 I \times I(B_i = \text{Southern}) I(Z_i = \text{No-take}).$$

$$\log(\mu_i) = \eta_i \Leftrightarrow \mu_i = \exp(\eta_i)$$

The reference group is the Northern open site. Its estimated expected coral trout count is

$$\hat{\mu}_{\text{Northern, Open}} = \exp(0.91629) \approx 2.50.$$

Under the log link, exponentiating a coefficient gives the corresponding **multiplicative effect on the expected count**.

For example, for the Central open site,

$$\exp(0.08516) \approx 1.089,$$

the expected count is about **1.089 times** the expected count in the reference group.

For the Southern open site,

$$\exp(1.12601) \approx 3.083,$$

the expected count is about **3.083 times** the expected count in the reference group.

For the Northern no-take site,

$$\exp(0.30421) \approx 1.356,$$

the expected count is about **1.356 times** the expected count in the reference group.

For the Central no-take site and Southern no-take site, we need to consider the interactive terms.

For the Central no-take site,

$$\exp(0.08516 + 0.30421 - 0.18870) \approx 1.222,$$

the expected count is about **1.222 times** the expected count in the reference group.

For the Southern no-take site,

$$\exp(1.12601 + 0.30421 + 0.40236) \approx 6.250,$$

the expected count is about **6.250 times** the expected count in the reference group.

## Part 5: Initial model comparison

We compare the interaction model with an additive model. The additive model assumes one common no-take/open expected-count ratio across latitude bands.

```
poisson_additive <- glm(
  coral_trout ~ latitude_band + zone,
  family = poisson,
  data = reef
)

anova(poisson_additive, poisson_full, test = "Chisq")
```

```
Analysis of Deviance Table

Model 1: coral_trout ~ latitude_band + zone
Model 2: coral_trout ~ latitude_band * zone
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       116     580.03
2       114     570.62  2   9.4029 0.009082 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
data.frame(
  model = c("Additive", "Interaction"),
```

```
  residual_df = c(
    df.residual(poisson_additive),
    df.residual(poisson_full)
  ),
  residual_deviance = c(
    deviance(poisson_additive),
    deviance(poisson_full)
  ),
  AIC = c(AIC(poisson_additive), AIC(poisson_full))
) |>
  knitr::kable(
    digits = 2,
    caption = "Initial Poisson model comparison."
  )
```

Initial Poisson model comparison.

| model | residual_df | residual_deviance | AIC |
|---|---|---|---|
| Additive | 116 | 580.03 | 944.7 |
| Interaction | 114 | 570.62 | 939.3 |

Under the Poisson assumptions, the likelihood-ratio statistic is 9.40 on 2 degrees of freedom, with $p = 0.0091$. The interaction model also has the smaller AIC (939.30 compared with 944.70). At this stage, both criteria favour retaining the interaction.

This conclusion is provisional because the Poisson model assumes $\mathrm{Var}(Y_i) = \mu_i$. The exploratory plots suggest that variability may be substantially greater.

## Part 6: Dispersion assessment

The full Poisson model has residual deviance 570.62 on 114 degrees of freedom. Its goodness-of-fit p-value is extremely small.

```
# Goodness-of-fit calculation used in the Gala class example
poisson_gof_p <- pchisq(
  deviance(poisson_full),
  df = df.residual(poisson_full),
  lower.tail = FALSE
)

n <- nrow(reef)
p <- length(coef(poisson_full))

# Pearson estimate used in the quasilikelihood class example
(phihat <- sum(residuals(poisson_full, type = "pearson")^2) /
    (n - p))
```

```
[1] 5.377568
```

```
c(
  goodness_of_fit_p = poisson_gof_p,
  pearson_dispersion = phihat
)
```

```
goodness_of_fit_p pearson_dispersion
     6.781645e-62       5.377568e+00
```

The Pearson dispersion estimate is approximately 5.38, far above the Poisson value of 1. The Poisson variance assumption is therefore not credible.

# Part 7: Overdispersion model and final selection

A quasi-Poisson model retains the log mean structure but uses

$$\mathrm{Var}(Y_i) = \phi \mu_i,$$

where $\phi$ is estimated from the data.

```
quasi_full <- glm(
  coral_trout ~ latitude_band * zone,
  family = quasipoisson,
  data = reef
)

quasi_additive <- glm(
  coral_trout ~ latitude_band + zone,
  family = quasipoisson,
  data = reef
)

anova(quasi_additive, quasi_full, test = "F")
```

```
Analysis of Deviance Table

Model 1: coral_trout ~ latitude_band + zone
Model 2: coral_trout ~ latitude_band * zone
  Resid. Df Resid. Dev Df Deviance      F Pr(>F)
1       116     580.03
2       114     570.62  2   9.4029 0.8743 0.4199
```

After scaling for overdispersion, the interaction comparison gives $F = 0.87$ with $p = 0.420$. There is insufficient evidence that the zone association differs among latitude bands. We therefore select the additive quasi-Poisson model.

We further assess the significance of each factor using the F-test.

```
quasi_latitude <- glm(
  coral_trout ~ latitude_band ,
  family = quasipoisson,
  data = reef
)

quasi_zone <- glm(
  coral_trout ~ zone ,
  family = quasipoisson,
  data = reef
)

anova(quasi_latitude, quasi_additive, test = "F")
```

```
Analysis of Deviance Table

Model 1: coral_trout ~ latitude_band
Model 2: coral_trout ~ latitude_band + zone
  Resid. Df Resid. Dev Df Deviance      F   Pr(>F)
1       117     639.15
2       116     580.03  1    59.13 10.941 0.001253 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(quasi_zone, quasi_additive, test = "F")
```

```
Analysis of Deviance Table

Model 1: coral_trout ~ zone
Model 2: coral_trout ~ latitude_band + zone
  Resid. Df Resid. Dev Df Deviance      F    Pr(>F)
1       118     918.47
2       116     580.03  2   338.44 31.313 1.336e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The small p-values indicate that both factors are statistically significant in the model.

The discussion below illustrates why the overdispersion check matters. The quasi-Poisson and Poisson fits have the same coefficient estimates for a given mean structure, but quasi-Poisson inflates the estimated covariance matrix to reflect the extra variability. Standard errors and confidence intervals become larger, and apparently significant terms may no longer be supported.

The calculation below follows the trout-egg class example. It shows that the coefficient estimates are unchanged while the quasi-Poisson standard errors are approximately the Poisson standard errors multiplied by $\sqrt{\widehat{\phi}}$.

```r
coef(poisson_additive)
```

```
         (Intercept)  latitude_bandCentral latitude_bandSouthern
           0.75788860           -0.01904819            1.37681562
         zoneNo-take
           0.56523235
```

```r
coef(quasi_additive)
```

```
         (Intercept)  latitude_bandCentral latitude_bandSouthern
           0.75788860           -0.01904819            1.37681562
         zoneNo-take
           0.56523235
```

```r
se_poisson <- coef(summary(poisson_additive))[, "Std. Error"]
se_quasi <- coef(summary(quasi_additive))[, "Std. Error"]

cbind(se_poisson, se_quasi, ratio = se_quasi / se_poisson)
```

```
                    se_poisson  se_quasi    ratio
(Intercept)          0.10825631 0.2516646 2.324711
latitude_bandCentral  0.13801910 0.3208545 2.324711
latitude_bandSouthern 0.10592269 0.2462396 2.324711
zoneNo-take           0.07497253 0.1742894 2.324711
```

```r
sqrt(summary(quasi_additive)$dispersion)
```

```
[1] 2.324711
```

Quasi-Poisson is not fitted by maximum likelihood. Consequently, AIC is unavailable, and ordinary unscaled likelihood-ratio tests are inappropriate. Nested quasi-Poisson models must be compared using the scaled F-test; `step()` and `drop1()` should not be used for this selection.

```r
summary(quasi_additive)
```

```
Call:
glm(formula = coral_trout ~ latitude_band + zone, family =
quasipoisson,
    data = reef)

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)            0.75789    0.25166   3.012  0.00319 **
latitude_bandCentral  -0.01905    0.32085  -0.059  0.95276
latitude_bandSouthern  1.37682    0.24624   5.591 1.52e-07 ***
zoneNo-take            0.56523    0.17429   3.243  0.00154 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 5.404279)

    Null deviance: 977.60  on 119  degrees of freedom
Residual deviance: 580.03  on 116  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5
```

The final mean model is

$$\log(\widehat{\mu}_i) = 0.758 - 0.019 I(B_i = \text{Central}) + 1.377 I(B_i = \text{Southern}) + 0.565 I(Z_i = \text{No-take}).$$

The estimated dispersion in this final model is 5.40.

## Part 8: Diagnostics and model alternatives

The following plots use the diagnostic quantities covered in the GLM lectures. With count data, horizontal banding in residual plots is expected because the response is discrete. The important questions are whether there is a systematic trend and whether individual sites are unusually extreme or influential.

```r
library(faraway)

old_par <- par(
  mfrow = c(2, 2),
  xpd = NA
)

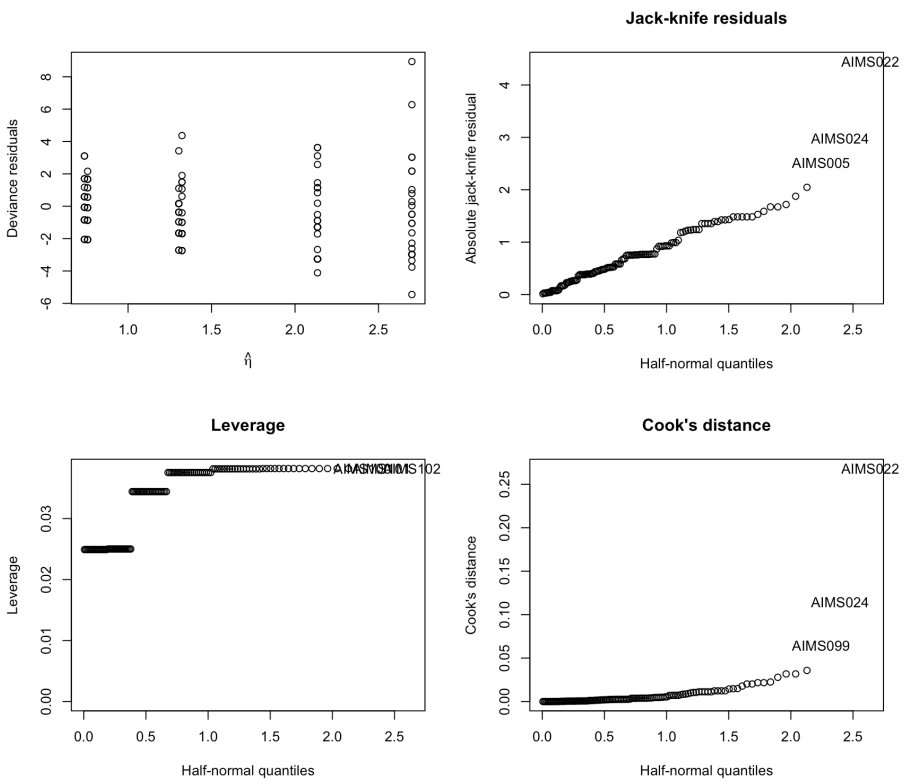
plot(
  residuals(quasi_additive) ~ predict(quasi_additive, type = "link"
  xlab = expression(hat(eta)),
  ylab = "Deviance residuals"
)

halfnorm(
  rstudent(quasi_additive),
  labs = reef$site,
  nlab = 3,
  ylab = "Absolute jack-knife residual"
)
title("Jack-knife residuals")

halfnorm(
  influence(quasi_additive)$hat,
  labs = reef$site,
  nlab = 3,
```

```
  ylab = "Leverage"
)
title("Leverage")

halfnorm(
  cooks.distance(quasi_additive),
  labs = reef$site,
  nlab = 3,
  ylab = "Cook's distance"
)
title("Cook's distance")
```




Diagnostics for the final additive quasi-Poisson model, following the class code. Labelled points are the three largest values in each half-normal plot.

```
par(old_par)
```

There is no strong systematic trend across the fitted linear predictor, although the residual spread is substantial. This is consistent with the overdispersion already accommodated by the quasi-Poisson model. Site `AIMS022` has a count of 61, the largest absolute jack-knife residual, and the largest Cook's distance. It is a Southern no-take observation.

```
reef[reef$site == "AIMS022", ]
```

```
      site latitude_band    zone coral_trout
22 AIMS022      Southern No-take          61
```

There is no evidence that `AIMS022` is a recording error, so it is retained. The residual plot does not indicate a systematic failure of the log-linear mean structure, and the interaction model has already been considered and was not supported after allowing for overdispersion.

To investigate its influence, we refit the same final model without `AIMS022` and compare the key estimates.

```r
row_022 <- which(reef$site == "AIMS022")

quasi_without_022 <- glm(
  coral_trout ~ latitude_band + zone,
  family = quasipoisson,
  data = reef,
  subset = -row_022
)

influence_comparison <- rbind(
  `All sites` = c(
    dispersion = summary(quasi_additive)$dispersion,
    zone_ratio = exp(coef(quasi_additive)["zoneNo-take"]),
    southern_ratio = exp(
      coef(quasi_additive)["latitude_bandSouthern"]
    )
  ),
  `Without AIMS022` = c(
    dispersion = summary(quasi_without_022)$dispersion,
    zone_ratio = exp(coef(quasi_without_022)["zoneNo-take"]),
    southern_ratio = exp(
      coef(quasi_without_022)["latitude_bandSouthern"]
    )
  )
)

knitr::kable(
  influence_comparison,
  digits = 3,
  caption = "Effect of excluding AIMS022 on key model estimates."
)
```

Effect of excluding AIMS022 on key model estimates.

| | dispersion | zone_ratio.zoneNo-take | southern_ratio.latitude_bandSouthern |
|---|---|---|---|
| All sites | 5.404 | 1.760 | 3.962 |
| Without AIMS022 | 4.376 | 1.588 | 3.623 |

Excluding `AIMS022` reduces the estimated dispersion from 5.40 to 4.38, so this site does not explain all the overdispersion. The no-take/open expected-count ratio falls from 1.76 to approximately 1.59, while the Southern/Northern ratio also becomes smaller. The directions and substantive pattern remain the same, although the precise effect sizes are influenced by this unusual observation. Among the model alternatives covered in the relevant classes, the additive quasi-Poisson model therefore remains the most appropriate choice. The influential site should nevertheless be mentioned when communicating the uncertainty and limitations of the analysis.

## Part 9: Estimates and uncertainty for the client

We first use `predict(..., type = "response")`, as in the class examples, to obtain the estimated site-level expected count for each latitude-band-by-zone combination.

```r
new_sites <- expand.grid(
  latitude_band = levels(reef$latitude_band),
  zone = levels(reef$zone)
)
```

```r
new_sites$expected_count <- predict(
  quasi_additive,
  newdata = new_sites,
  type = "response"
)

knitr::kable(
  new_sites,
  digits = 2,
  caption = "Model-estimated site-level expected coral trout counts
)
```

Model-estimated site-level expected coral trout counts.

| latitude_band | zone | expected_count |
|---|---|---|
| Northern | Open | 2.13 |
| Central | Open | 2.09 |
| Southern | Open | 8.45 |
| Northern | No-take | 3.76 |
| Central | No-take | 3.68 |
| Southern | No-take | 14.88 |

Under the additive model, no-take and open counts have the same estimated ratio in every latitude band. We obtain the interval directly from the `zoneNo-take` coefficient.

```r
zone_estimate <- coef(quasi_additive)["zoneNo-take"]
zone_se <- sqrt(vcov(quasi_additive)["zoneNo-take", "zoneNo-take"])

critical_value <- qt(
  0.975,
  df = df.residual(quasi_additive)
)

zone_log_interval <- zone_estimate +
  c(-1, 0, 1) * critical_value * zone_se

zone_ratio <- exp(zone_log_interval)
names(zone_ratio) <- c("Lower 95%", "Estimate", "Upper 95%")
zone_ratio
```

```
Lower 95%  Estimate Upper 95%
 1.246114  1.759857  2.485402
```

After accounting for latitude band, the expected coral trout count at a no-take site is estimated to be **1.76 times** that at an open site, with a 95% confidence interval from **1.25 to 2.49**. Equivalently, the model estimates about 76% more recorded coral trout at no-take sites, although plausible values range from about 25% more to about 149% more.

The model also estimates substantially larger expected counts in the Southern band. Holding zone fixed, the estimated Southern/Northern ratio is $\exp(1.377) = 3.96$. The Central and Northern bands have very similar estimated counts.

There is insufficient evidence, after allowing for overdispersion, that the no-take/open association differs by latitude band. The common-ratio additive model is therefore an adequate and more parsimonious summary for this report.